FTC INTRO ROBOTICS CAMP

A week-long summer camp where middle schoolers form mini teams and design, build, program, and present an FTC-scale robot.

GRADES	LENGTH	GROUP SIZE	EXPERIENCE
Rising 6th to 9th	5 days, about 3 hours a day	5 to 8 per robot; two robots for 9 to 16	None needed

This week-long summer camp puts middle schoolers through the real process of building an FTC-scale robot. Campers split into one or two mini teams, and each team designs, builds, programs, and promotes its own robot with step-by-step help from experienced FIRST Tech Challenge students and mentors.

Campers work on electronics, Java programming, mechanical design, and the troubleshooting that fills most of a real FTC season. They also do the parts people forget about: naming and branding a team, splitting up roles, and presenting their work. The week ends with a game-day test on a practice field and a showcase for families.

It's also a recruiting pipeline. A good number of campers go on to join or start an FTC team, and the host team gets outreach hours, photos, and a story for its engineering portfolio.

WE PROVIDE

- Daily lesson plans for all five days
- Training for your mentors before camp
- Starter code, build guides, and a game-day score sheet
- Printed team materials: posters, stickers, name tags
- A Spark Gears mentor on call all week

THE HOST PROVIDES

- A room with tables, outlets, and space for a practice field
- Campers, sign-ups, and marketing
- One FTC robot kit per mini team (a past season robot or a starter kit is fine)
- Mentors from your team for every day of camp

CAMPERS BRING

- A laptop: personal or school issued
- Closed-toe shoes
- A snack and a water bottle

No robot kit? Tell us when you request support. We'll help you find one to borrow or plan a smaller version of the camp around what you have.

A TYPICAL DAY

Every day follows the same rhythm so campers know what to expect and mentors can plan.

BLOCK	TIME	WHAT HAPPENS
Warm-up	15 min	Quick challenge or question of the day, attendance, safety glasses on
Mini-lesson	20 to 30 min	One new idea, taught with a demo on a real robot
Team work	75 to 90 min	Teams build, wire, or code with mentors rotating between them
Break	15 min	Snack, water, away from the tables
Team work	30 to 45 min	Keep going, test, fix what broke
Stand-up	10 min	Each team says what they finished, what's stuck, and the plan for tomorrow

DAY BY DAY

DAY 1: MEET THE CHALLENGE

BLOCK	ACTIVITY	MENTOR NOTES
Morning	Introductions, camp rules, safety glasses. Watch the current FTC game reveal video together.	Keep the video to about 5 minutes. Pause to point out scoring.
Mid	Break into mini teams. Each team picks a name, a number, and draws a logo.	Branding gets shy campers talking. Have markers and poster paper ready.
Late	Tour a real FTC robot. Campers find the Control Hub, motors, servos, and battery.	Let them hold parts. Let them drive it with a gamepad for a few minutes.
Wrap	Each team picks one game task to focus on this week.	Smaller goals finish. One task done well beats three half done.

DAY 2: DESIGN AND START BUILDING

BLOCK	ACTIVITY	MENTOR NOTES
Mini-lesson	How drivetrains work, how to read a build guide, using hex keys and nuts safely.	Show a finished drivetrain next to its parts.
Team work	Sketch the robot on paper. Decide who builds the drivetrain and who builds the mechanism.	Every camper gets a job. Rotate jobs each day.
Build	Start the drivetrain: frame, motors, wheels.	Check that motors are mounted the same way on both sides.

DAY 3: KEEP BUILDING

BLOCK	ACTIVITY	MENTOR NOTES
Mini-lesson	Wiring: Control Hub ports, motor and servo cables, keeping wires tidy.	Show a messy wiring job next to a clean one.
Build	Finish the drivetrain, wire it, start the scoring mechanism (intake, arm, or claw).	Cable management matters. Loose wires cause most robot failures.
Test	First drive test with sample code.	Tape a box on the floor and have them drive in and out of it.

DAY 4: PROGRAM THE ROBOT

BLOCK	ACTIVITY	MENTOR NOTES
Mini-lesson	Java in OnBot Java: what an OpMode is, gamepad input, setting motor power.	OnBot Java runs in the browser from the Control Hub, nothing to install.
Code	Write a TeleOp so a driver can control the robot, then map the mechanism to buttons.	Start from the code below. Let campers change values and see what happens.
Stretch	A 5 second autonomous that drives forward and parks.	Timing based is fine. Encoders are a bonus.

DAY 5: GAME DAY AND SHOWCASE

BLOCK	ACTIVITY	MENTOR NOTES
Morning	Final fixes and driver practice.	Freeze the code an hour before the game. No last-second changes.
Game test	Each team plays timed matches on the practice field and records scores.	Use the score sheet. Cheer loudly.
Showcase	Families come in. Each team presents its robot, team name, and what they learned.	2 to 3 minutes per team. Every camper says one thing.

STARTER TELEOP CODE

Paste this into OnBot Java on the Control Hub. In the robot configuration, name the drive motors left and right. Campers then add their own mechanism controls inside the loop.

CAMPDRIVE.JAVA

```
package org.firstinspires.ftc.teamcode;

import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
import com.qualcomm.robotcore.hardware.DcMotor;

@TeleOp(name = "Camp Drive")
public class CampDrive extends LinearOpMode {
    @Override
    public void runOpMode() {
        DcMotor left = hardwareMap.get(DcMotor.class, "left");
        DcMotor right = hardwareMap.get(DcMotor.class, "right");
        right.setDirection(DcMotor.Direction.REVERSE);

        waitForStart();

        while (opModeIsActive()) {
            double drive = -gamepad1.left_stick_y;   // push forward = drive forward
            double turn = gamepad1.right_stick_x;

            left.setPower(drive + turn);
            right.setPower(drive - turn);

            telemetry.addData("Drive", drive);
            telemetry.addData("Turn", turn);
            telemetry.update();
        }
    }
}
```

TEAM ROLES

Each camper takes one role per day and rotates, so everyone builds, codes, and drives at least once.

ROLE	WHAT THEY DO
Builder	Assembles and repairs the robot, keeps track of parts
Programmer	Writes and tests code, works with the driver to tune controls
Driver	Practices driving, gives feedback on what's hard to control
Team captain	Keeps the team on schedule, leads the daily stand-up
Brand lead	Logo, team poster, photos, and the showcase presentation

LOGISTICS AND SAFETY

Target grades	Rising 6th through 9th graders. Strong 5th graders can join with a parent's okay.
Group size	About 5 to 8 campers per robot. With two robots, run two teams of up to 8.
Daily length	About 3 hours a day for 5 days. Mornings work best in summer.
Mentors	At least 1 experienced mentor per team, plus 1 adult on site at all times.
Safety	Safety glasses for every camper and mentor whenever a robot is powered or being built. Closed-toe shoes. Robots are disabled when anyone reaches into them. Batteries are unplugged when not in use and never charged unattended.
Families	Invite families to the Day 5 showcase. It's the best photo opportunity of the week.

QUESTIONS? TALK TO A HUMAN.

Email sparkgearsinitiative@gmail.com or call 571-429-9220. Request a mentor and a supply kit at sparkgearsinitiative.org/for-teams.html. We usually reply within two school days.